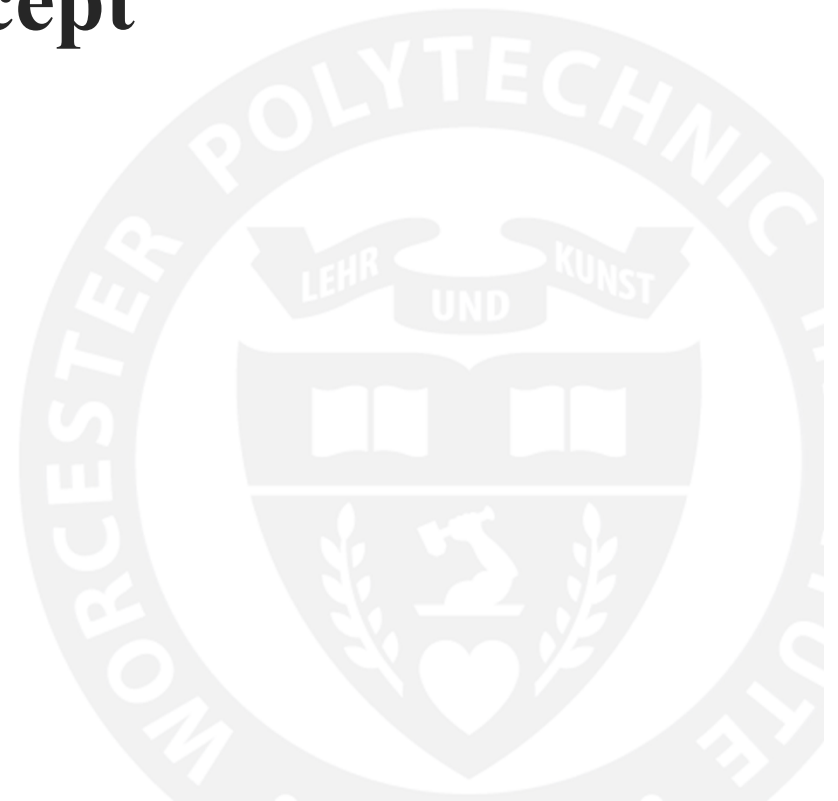




WPI

Lecture 2a - Basic Concept



Basic Concept

- Data Set = { Instance, or Sample }

- Attribute, Feature

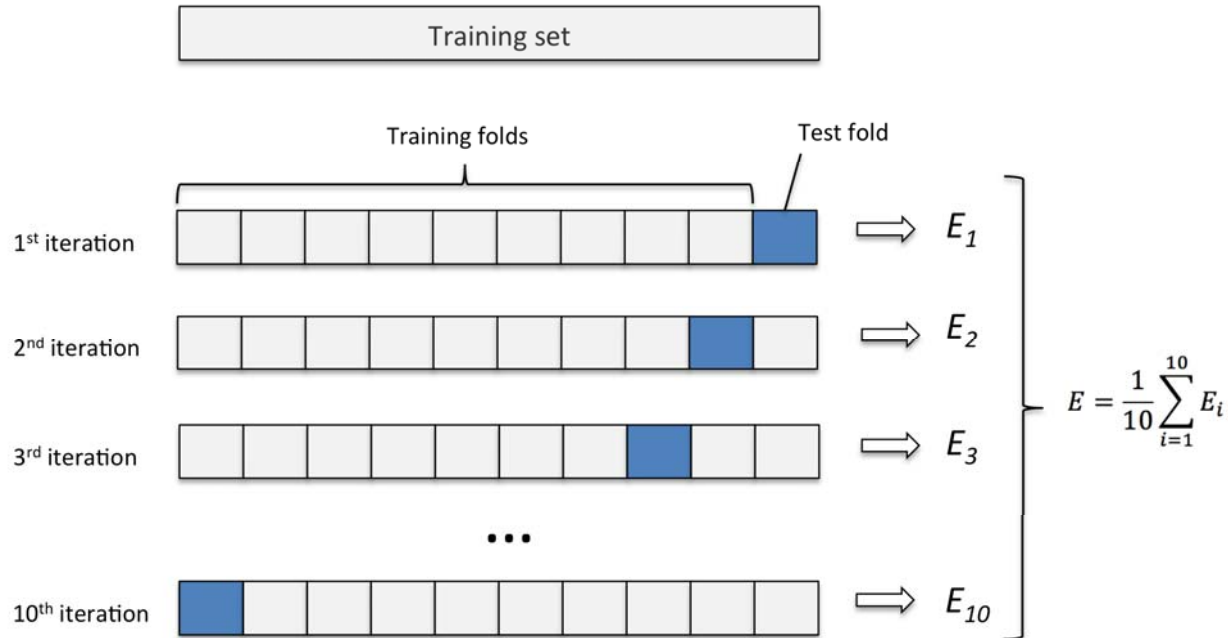
$$D = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$$

$$\mathbf{x}_i = (x_{i1}; x_{i2}; \dots; x_{id}) \quad d - \text{Dimensionality}$$

- Training data & testing data
- Classification
 - Discrete values, such as *binary classification*, *multi-class classification*
 - Positive class, negative class

Training Error and Validation

- Error rate = 1 – accuracy
- Training set, testing set
 - Simple “hold-out”: 70% samples for training, 30% samples for testing
 - Cross Validation: k -fold cross validation



Bootstrapping

- Bootstrap sampling

- Given a data set D with m samples, randomly select one each time and put it into training set D'
- The probability of samples not being selected into D' is:

$$\lim_{m \rightarrow \infty} \left(1 - \frac{1}{m}\right)^m \mapsto \frac{1}{e} \approx 0.368$$

- We still use m samples for training, but about 1/3 of original dataset not included in D' can be used for testing.

Performance Measurement

- Given a dataset $D = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_m, y_m)\}$
- Mean square error (MSE)

$$E(f; D) = \frac{1}{m} \sum_{i=1}^m (f(\mathbf{x}_i) - y_i)^2$$

- Error Rate

$$E(f; D) = \frac{1}{m} \sum_{i=1}^m \mathbb{I}(f(\mathbf{x}_i) \neq y_i)$$

- Accuracy

$$\begin{aligned} \text{acc}(f; D) &= \frac{1}{m} \sum_{i=1}^m \mathbb{I}(f(\mathbf{x}_i) = y_i) \\ &= 1 - E(f; D) . \end{aligned}$$

Confusion Matrix

		Prediction outcome		
		positive	negative	
Actual value	positive	TP	FN	$TP + FN$
	negative	FP	TN	$FP + TN$
		$TP + FP$	$FN + TN$	

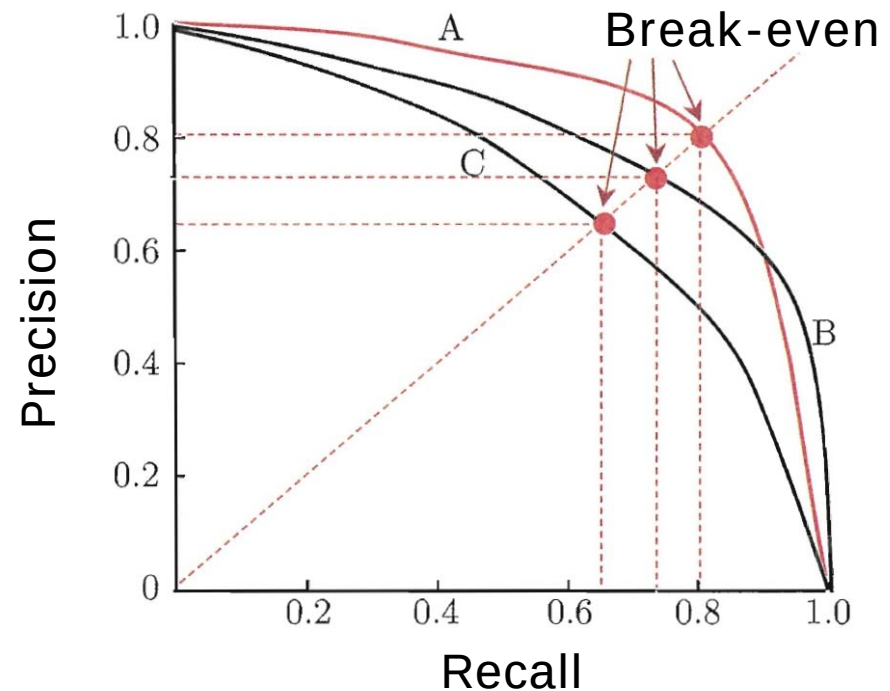
- **Accuracy** measures overall accuracy of the model classification.
 - $\text{Accuracy} = (TP + TN) / (TP + FP + TN + FN)$
 - **Error Rate** = $(FP + FN) / (TP + FP + TN + FN) = 1 - \text{Accuracy}$

Precision and Recall

- **Precision** measure accuracy of a class, when predicts “yes” how often it is correct.
 - $\text{Precision} = \text{TP}/(\text{TP}+\text{FP})$
- **Recall**: When it is actually “yes” how often it predicts “yes”.
 - $\text{Recall} = \text{TP}/(\text{TP}+\text{FN})$
- **Sensitivity**: (true positive rate) is the ratio of the number of correct positive samples to the number classified as positive
 - $\text{Sensitivity} = \text{TP}/(\text{TP}+\text{FN})$
- **Specificity** is the same ratio for negative samples
 - $\text{Specificity} = \text{TN}/(\text{TN}+\text{FP})$

Precision-Recall Curve

- Area of P-R curve is often used as an indicator for classification performance.



F1 Score

- F1 score is the harmonic mean of Precision and Recall

$$F1 = \frac{2 \times P \times R}{P + R} \qquad \frac{1}{F1} = \frac{1}{2} \cdot \left(\frac{1}{P} + \frac{1}{R} \right)$$

- In general,

$$F_{\beta} = \frac{(1 + \beta^2) \times P \times R}{(\beta^2 \times P) + R} \qquad \frac{1}{F_{\beta}} = \frac{1}{1 + \beta^2} \cdot \left(\frac{1}{P} + \frac{\beta^2}{R} \right)$$

- When $\beta > 1$, recall has more impact;
- When $\beta < 1$, precision has more impact.

ROC Curve

- Receiver Operating Characteristics

- True positive rate

$$TPR = \frac{TP}{TP + FN} ,$$

- False positive rate

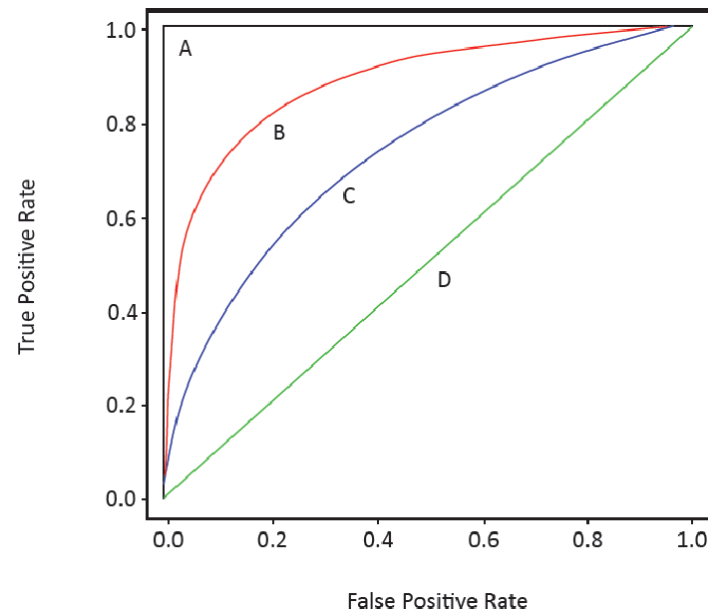
$$FPR = \frac{FP}{TN + FP} .$$

- AUC: area under ROC curve

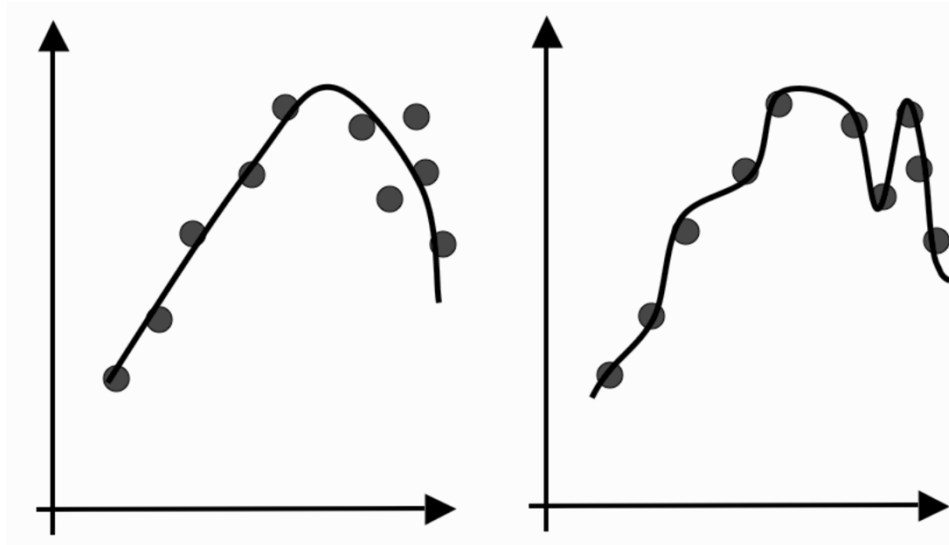
- A: perfect with AUC 1

- D: random classification AUC 0.5

- B is better than C



Overfitting



- (Left) finding the generating function
- (Right) matching the inputs perfectly, including the noise in them (on the right).
- Overfitting reduces the generalization capabilities of the model.